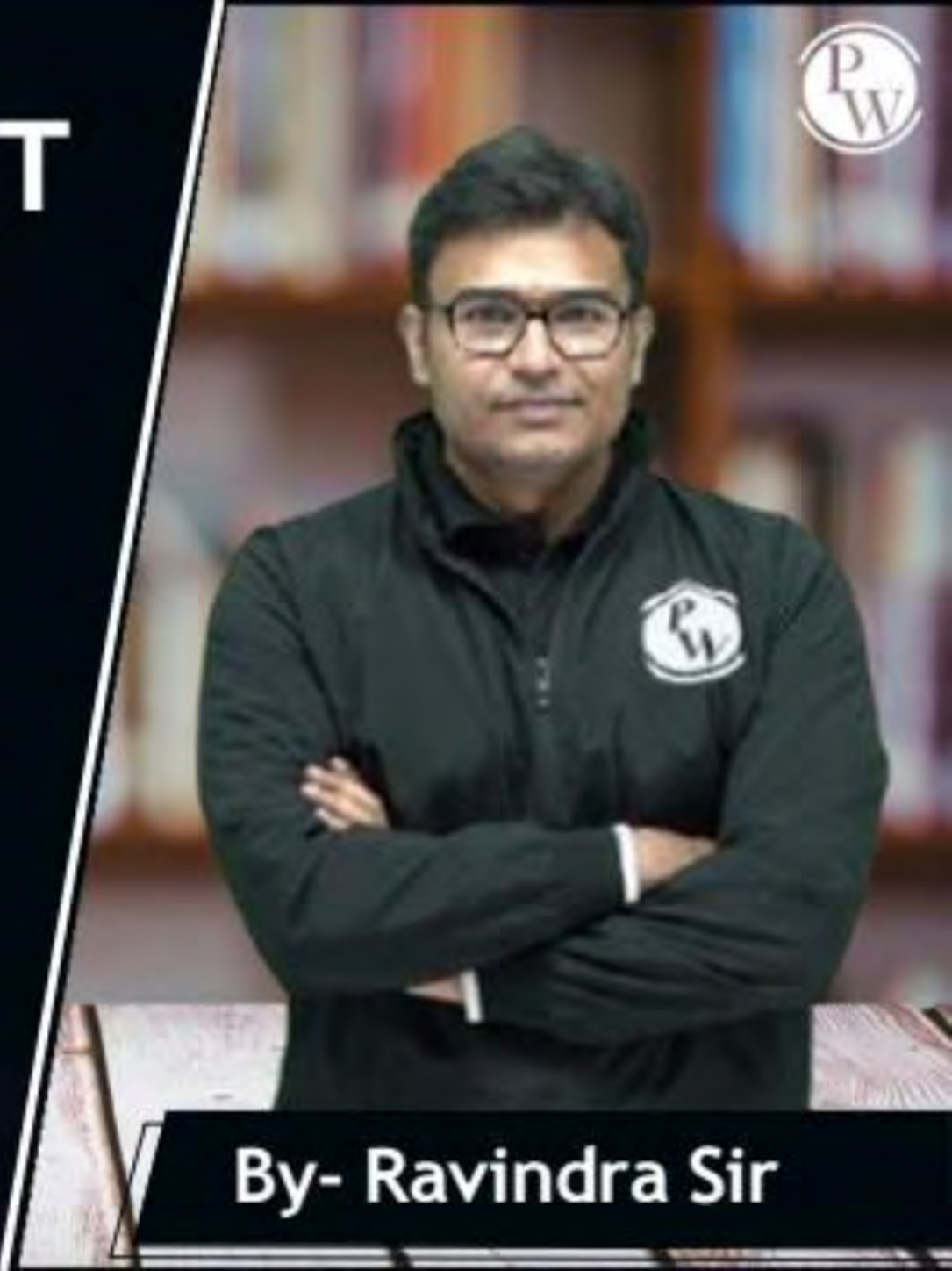
Computer Science & IT

COMPUTER NETWORK
(CN)

ISO-OSI Stack

Lecture No. 2



By- Ravindra Sir

Recap of Previous Lecture



Topic

Topic

Topic

Part 1

Topics to be Covered



- Topic
- Topic
- Topic

ISO OSI Part 2

Inspiring Stories : Chandu Singh Nagar



Background: From a village near Delhi.

Struggles: Left home to fight with Netaji Bose's army.

Education: Learned discipline in Azad Hind Fauj.

Achievements: Fought in Southeast Asia, jailed, returned a hero.

Impact: Showed even small villages had global fighters.

Inspiring Stories : Dariyav Singh Nagar



Background: From Greater Noida region.

Struggles: Saw British humiliating locals.

Education: Learned from local elders to fight back.

Achievements: Forced British to work in fields, later executed.

Impact: Still remembered as a symbol of pride and defiance.

Inspiring Stories : Umabai Kundapur

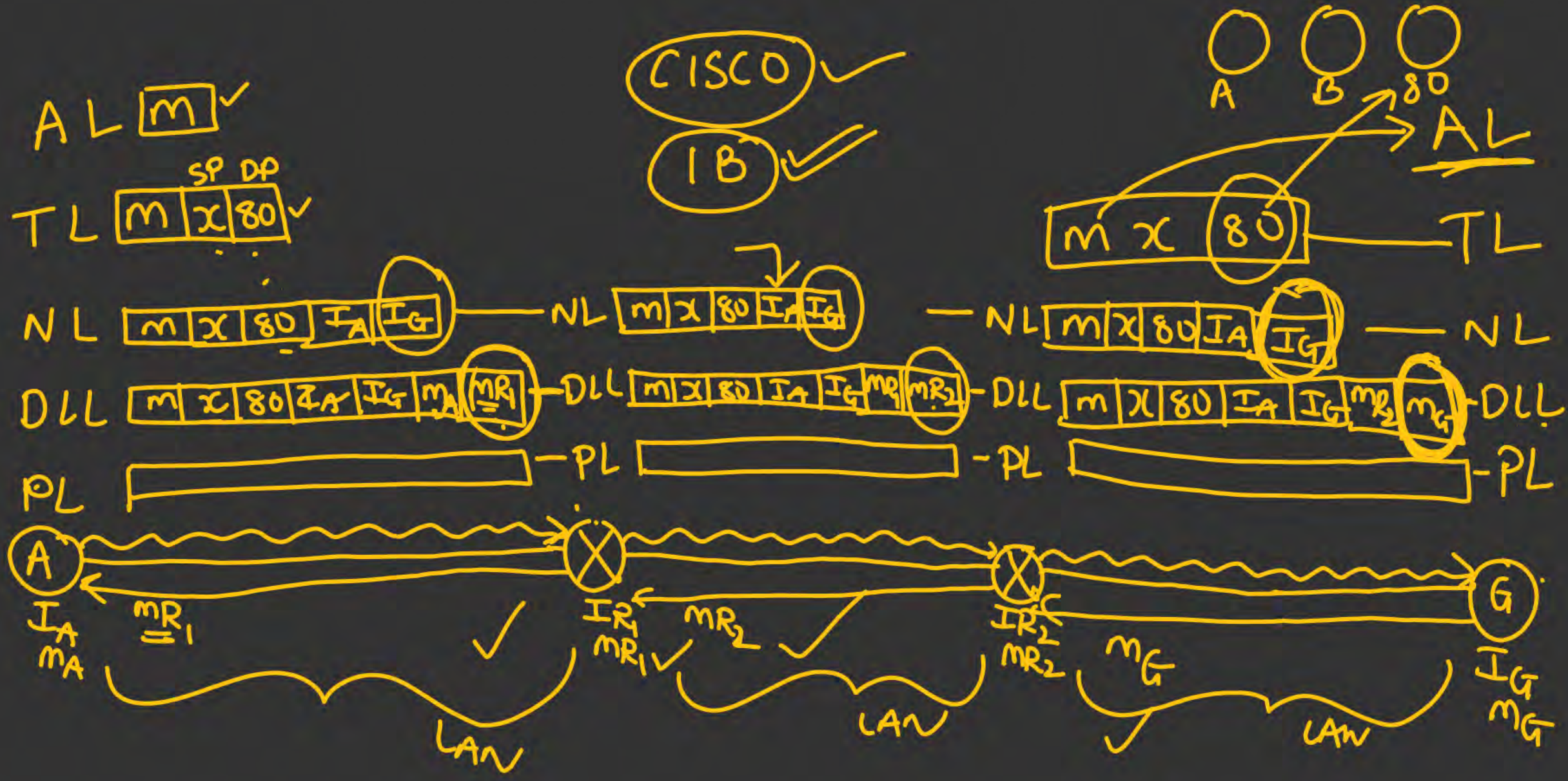


Background: Born in 1892 in Hubli, Karnataka. She became part of the freedom movement.

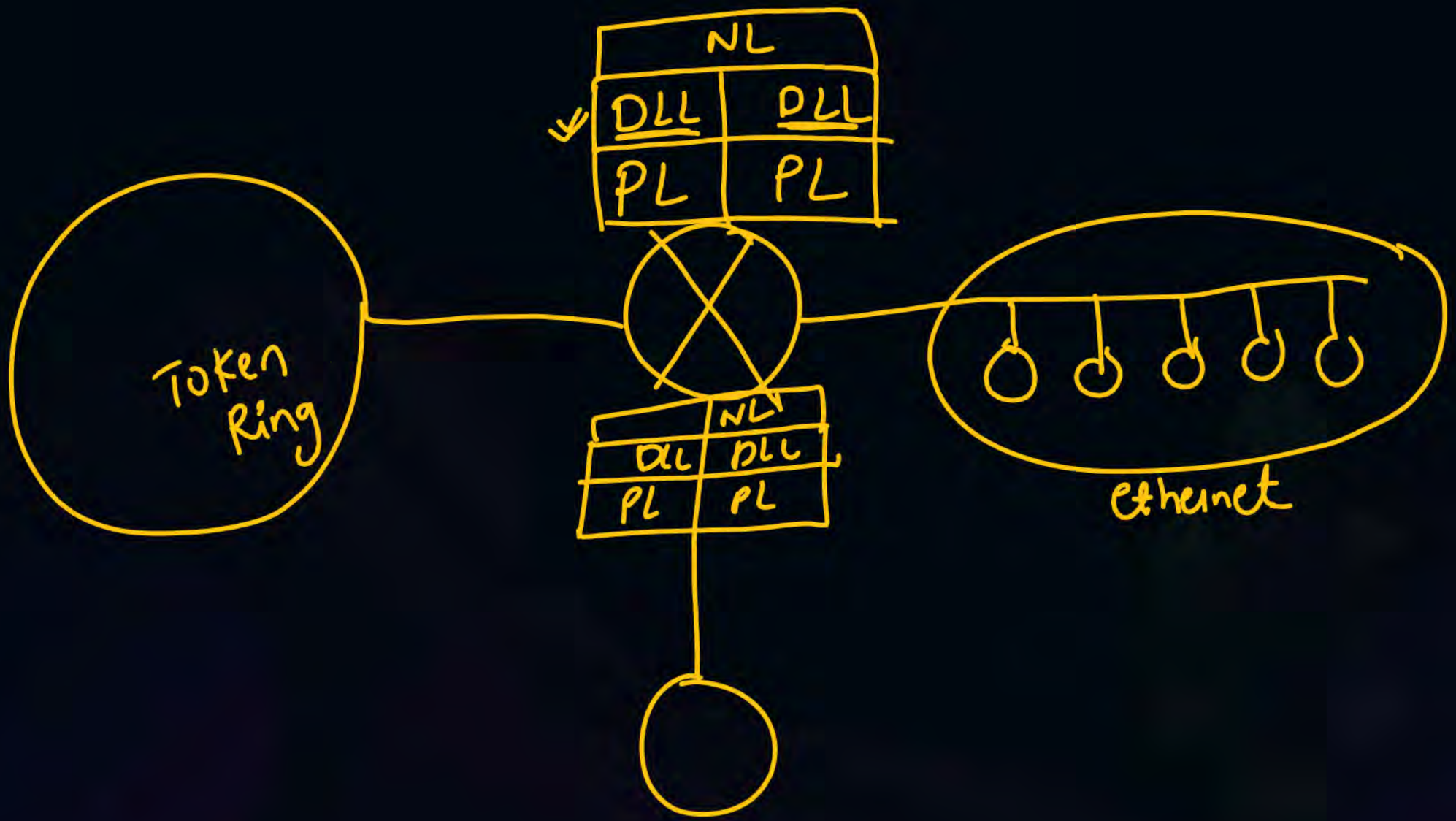
Struggles: She lived in a time when women were not taken seriously in public life.

Achievements: Umabai led the women's wing of Seva Dal (a volunteer group). She drew women into the freedom fight, led plays to inspire people, and was arrested and jailed in 1932.

Impact: She showed that even when women had no power, they could lead and inspire.


AL
m
SP DP
TL
m x 80
CISCO
IB
A
B
80
AL
m x 80
TL
NL
m x 80 IA IG
DLL
m x 80 IA IG mA mR1
m x 80 IA IG mR2 mR2
m x 80 IA IG mR2 mG
PL
A
IA
mA
mR1
LAN
IR1
mR1
mR2
LAN
IR2
mR2
mG
LAN
G
IG
mG


NL
IP
(DG, DIP)
ARP
DLL
frame

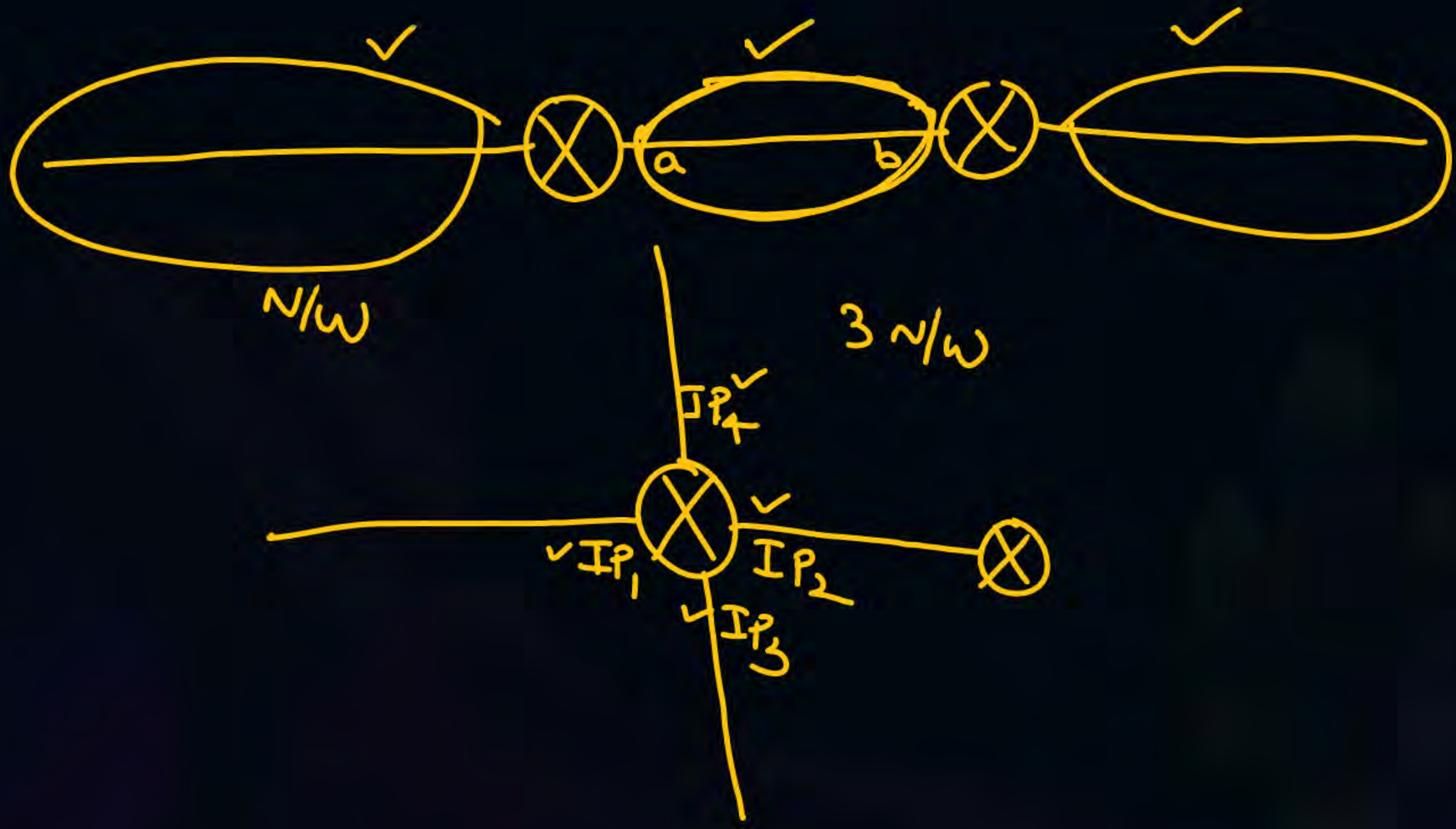

NL
DLL
DLL
PL
PL
Token Ring
NL
DLL
DLL
PL
PL
Ethernet

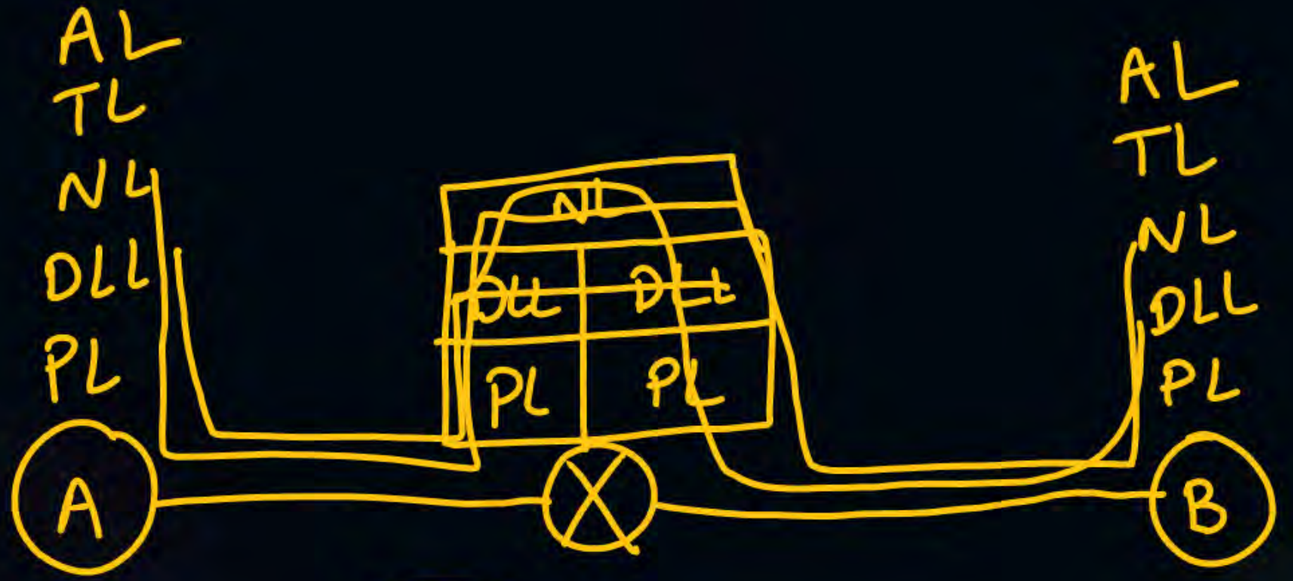


Every network is bounded by Router.

Broadcasting Done at DLL, Cannot cross a router




a
b
N/W
3 N/W
I_{P_4}
I_{P_1}
I_{P_2}
I_{P_3}



AL-2
TL-2
NL-3
DLL-4
PL-4


AL
PL
SL
X
TL
NL
DLL
PL

Not imp for gate :-

Session layer :-

1) authentication and authorization

→ Passwords

2) check pointing



3) Synchronization

Audio + video ✓

4) dialog control



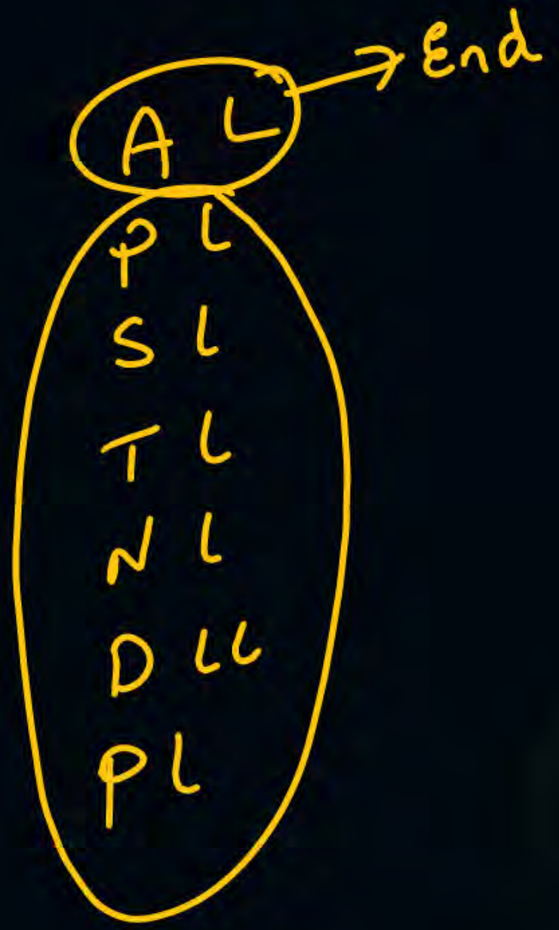
5) logical grouping of operations

Atomicity → DBMS

Not imp for Gate :

Presentation layer :

1) Character translation
   ASCII → EBCDIC
2) Encryption and decryption
3) Compression and expansion



Ethernet (IEEE 802.3)

1) Topology bus

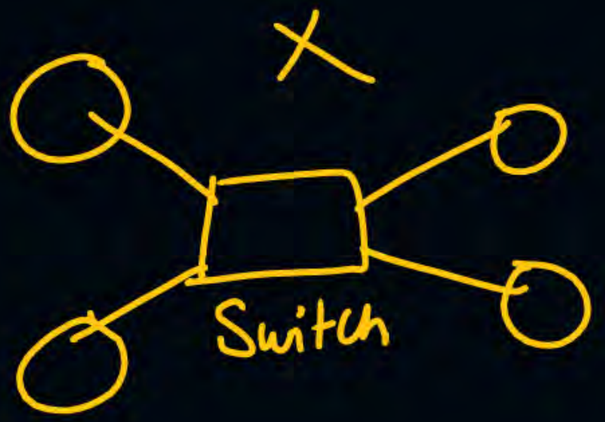


2) Access Control - CSMA/CD

3) No ACK

4) Data Rates : 10mbps, 100mps, 1Gbps

- 100mps → fast ethernet
- 1Gbps → Gigabit ethernet



5) Encoding: manchester encoding

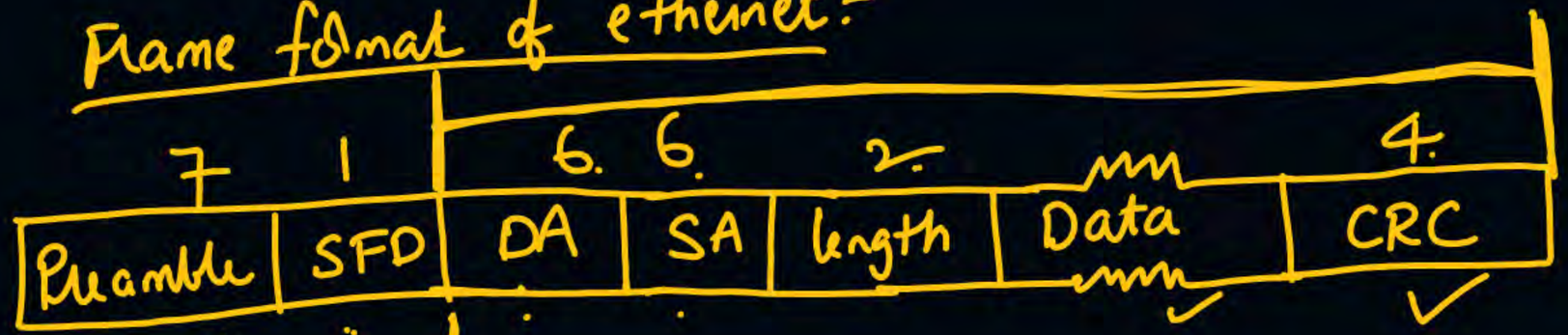


AL — message ✓
TL — Segment ✓
NL — Datagram ✓
DLL — Frame ✓
PL — 1 PDU ✓

LANs work at DLL

Ethernet is used at DLL

Frame format of ethernet:-



added by PL

DA, SA → MAC

16 bits

2^{16}

CRC - 32 bit CRC

Preamble :- It is used to indicate start of the frame and also for Synchronization

Preamble - 7B - 10101010.....10

SFD - 1B - 10101011

DA, SA → MAC addresses

Types of MAC

(i) unicast address → LSB of 1 byte is '0'

1A : 2B : 3C : 4D : 5A : 6A

⇓

00011010

unicast add

↓

We can allot the address only to one station

multicast address:-

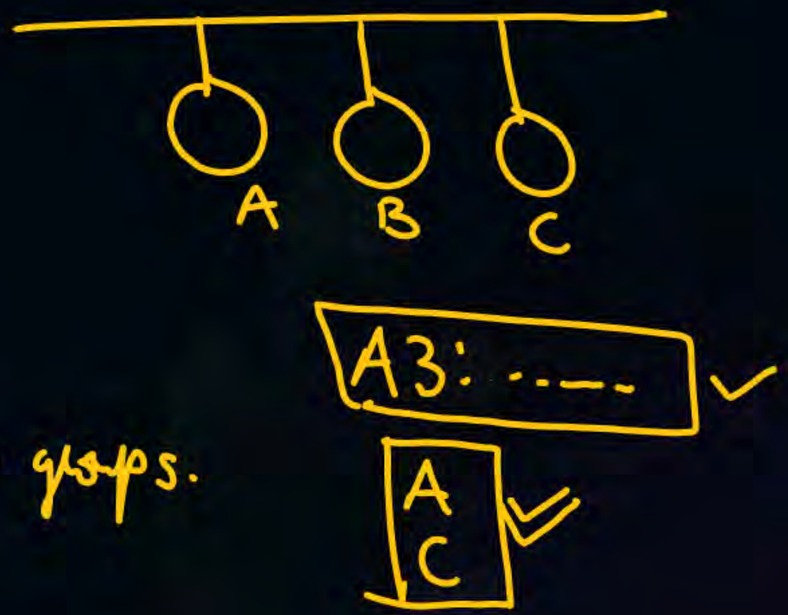
If the LSB of 1 byte is '1', then it is called multicast add.

Ex. A3 : B1 : C2 : 13 : 56 : 98



multicast add

We assign multicast add to groups.



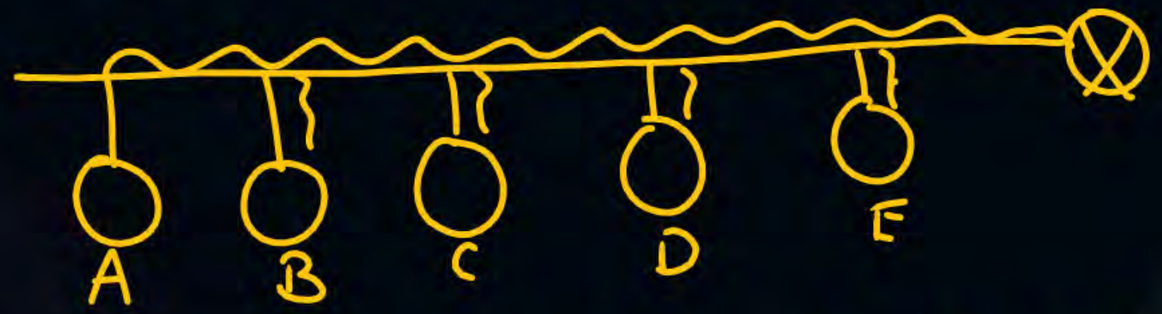
Broadcast address :-

If all the bits are '1', then it is called broadcast address

11111111.11111111.11111111.11111111 → FF:FF:FF:FF

LBA → IP → 255.255.255.255 (DIP)

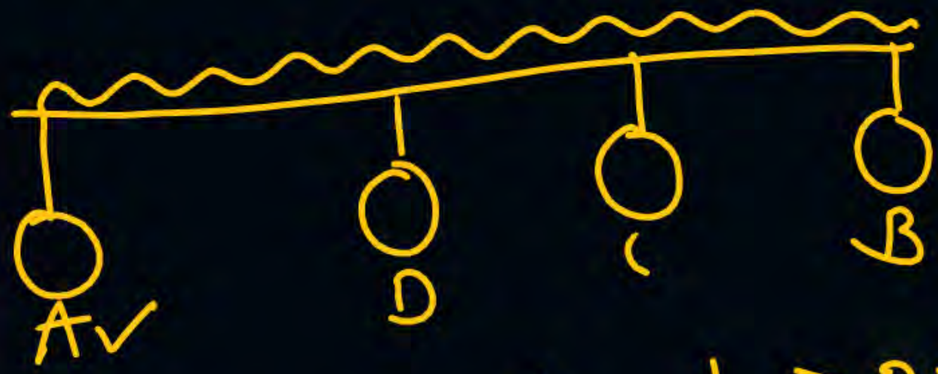
FF:FF:FF:FF:FFFF (DMac)



DMAC: FF.FF.FF.FF:FF.FF

Length :- (16 bits)

2^{16} maximum length → monopolization



L_{max} = 1500 B
Data

$L \geq 2 * T_p * B$

$L_{min} \geq 64$
frame

	min	max
Data	46B	1500B
frame	64B	1518B

✓

CRC: 32 bit

→ Why is CRC added at the end?

NIC (Network interface card) Calculates the CRC.



CRC is Calculated when the packet is being transmitted

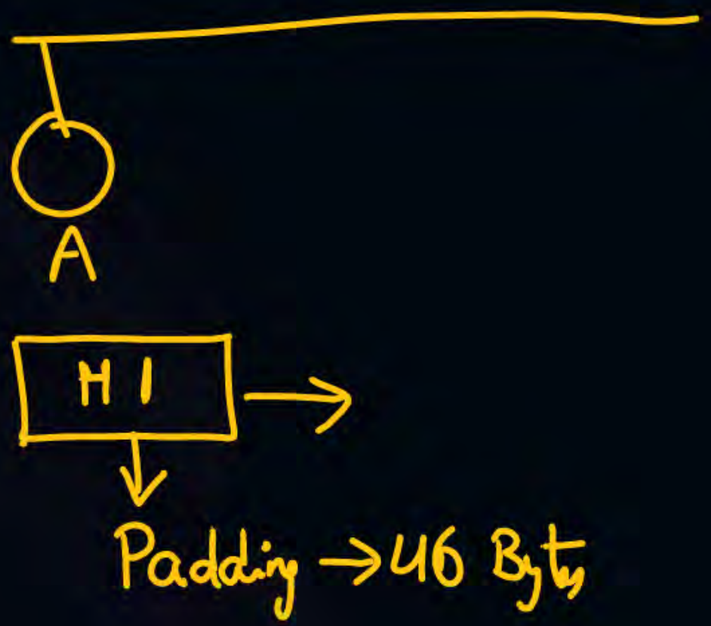
So CRC is added at the end

Disadvantages of ethernet

(i) Not applicable fd RT applications, because of collisions



(ii) Not applicable for interactive applications



(iii) There are no priorities in ethernet.
So it is not applicable for client server application

THANK - YOU